Sheet 1

1 -.draw a flowchart Print Hello World 10 times.

2- . draw a flowchart thats Find the sum of 5 numbers.

3-. draw a flowchart thats Add 10 and 20 and put it in sum.

4- draw a flowchart to determine the state of a student (succeeded of failed) depending on his mark in the exam, full .make is assumed to bo 100.

5- convert this numbers

** 1011101010 from B to H * FFEA from H to B*

** 76 fron D to B * 84 from D to H*

6- The equivalent resistance of resistors connected in series is calculated by adding the resistances of individual resistors. Req = R1 + R2

The formula for resistors connected in parallel is a little more complex. Given two resistors with resistances R1 and R2 connected in parallel the equivalent resistance

.is given by the inverse of the sum of the inverses

Req = 1/R1 + 1/ R2/1

Write a c++ program that given 2 resistances (R1,R2) outputs the equivalent resistance (Req) when they are connected in .series and when they are connected in parallel

:Sample Ran 1: (inputs in Green, outputs in Black)

Green 1

Green 1

Black 2.0

Black 0.5

7- Write a C++ program Count Change to count change. Given _ ,the number of quarter, dimes

nickels and cents the program should output the total as a .single value in dollars and cents

:Hint

One dollar corresponds to 100 cents ● One quarter ● corresponds to 25 cents

One dime corresponds to 10 cents ●

One nickel corresponds to 5 cents ●

For example if we have: 3 quarters, 2 dimes, 1 nickel and 6 .cents, the total is 1.06 dollars

8 -The Pythagorean Theorem states that the sum of the squares of the two sides of the right angle triangle is equal to the square .of the hypotenuse

For example: 3, 4 and 5 are sides of the right angle triangle as (32 + 42 = 52) they form a Pythagorean Triple

,Given 2 numbers, m and n where m >= n

a Pythagorean Triple can be generated by the following .formulae

a = m^2 – n^2

b = 2mn

c = root (a^2 + b ^ 2)

Write a C++ program that prints the values of the Pythagorean Triple generated by the formulae above, given m .and n

Hint: To calculate the square root and the power, use the class Math which contains the methods Math.sqrt(x) and Math.pow(x,n)

:Sample Ran 1: (inputs in Green, outputs in Black)

Green 2

Green 1

3.0

4.0

5.0

9- Write a C++ program that for a given number of apples, tells the user how many dozens of apples(s) he has and how many .extra apples are left over

For Example: if the number of apples = 40, the output should be: 3 dozens and 4 apples

:Sample Ran 1: (inputs in Green, outputs in Black)

40

3

4

:Sample Ran 2: (inputs in Green, outputs in Black)

1

0

1